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A multiphase model for fluid–particle flows with added mass and phase change

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ABSTRACT

In our prior work, a kinetic-based model for fluid–particle flows with added mass was derived (Fox et al., 2020) and successfully validated for non-reacting cases without mass transfer (Boniou and Fox, 2023). In this work, the multiphase model is extended to allow for the added-mass phase to have a different temperature and density than the particle and bulk-fluid phases, and to account for mass transfer from the particle phase to the added-mass phase. For this purpose, mass and energy transfer between particles and their added mass, and the added-mass and bulk-fluid phases are included. As before, the particle-phase density ρ_p is assumed constant, but now the added-mass density ρ_a is allowed to differ from the bulk-fluid density ρ_f. These densities are coupled through the assumption that their pressures are equal, i.e., $p_a = p_f$, and determined by a fluid-phase equation of state. Since mass transfer will change the particle volume, an additional balance equation for the particle number density N_d is required to complete the multiphase model. The computational algorithm for the extended model is tested using sublimation and condensation of carbon-dioxide particles interacting with a strong shock wave.

1. Introduction

Dispersed multiphase flows in the presence of a particle phase with heat and mass transfer occur in numerous natural and industrial problems such as volcanic ash plumes [1], dust explosions [2], rocket propulsion [3,4], cooling of electronic components [5,6], cooling in space environments [7], greenhouse gas control [8], and biofuel combustion [9]. The complexity of high-speed multiphase flows stems from the interaction between shock waves and the particles [10], which highly depends on the volume fraction [11]. While dilute mixtures can be approximated with experiments and numerical simulations of isolated particles subject to a shock, the presence of neighboring particles cannot be neglected when their volume fraction increases [12]. In denser mixtures, a particle is not only subject to the incident shock wave, but also to shock reflections and expansion waves coming from the other particles and compaction waves transmitted through the granular phase. When the mixture becomes very dense, the particles are always in contact and the inter-particle interactions become dominant. When mass transfer between the particles and gas phase is present, the situation is further complicated by the change in particle size and the fluid-phase pressure. When exothermal chemical reactions are included, the temperature and chemical composition in the fluid phase near the particle surface can be very different than in the bulk fluid. Predictive modeling of gas–particle flows requires to handle all possible regimes with an appropriate physical description.

In our prior work [13–15], an added-mass model (referred to here as the 9-equation model) was developed for particles without mass transfer or chemical reactions following the seminal work of [16]. The basic idea is to extend the classical two-fluid model for fluid–particle flows to include a third phase representing the fluid surrounding a particle. The full three-fluid model for bulk fluid, particles, and added-mass has mass, momentum, and energy equations for each of the three phases [16]. The volume fractions of the three phases, denoted, respectively, by $\alpha_f^\star$, α_p, and α_a, sum to one. By forcing the added-mass and particle phases to move with the same velocity (i.e., $\mathbf{u}_a = \mathbf{u}_p$), and the added-mass and fluid phases to have the same pressure (i.e., $p_a = p_f$), the virtual-mass force appearing in the classical two-fluid model is obtained [13,16], and the momentum balance for the added-mass phase can be removed. The added-mass phase requires a mass balance (i.e., for $\rho_a\alpha_a$), but otherwise the mathematical form of the multiphase model remains unchanged. In the absence of mass transfer between the particle and fluid phases and of homogeneous chemical reactions, the added-mass phase can safely be assumed to have the same temperature

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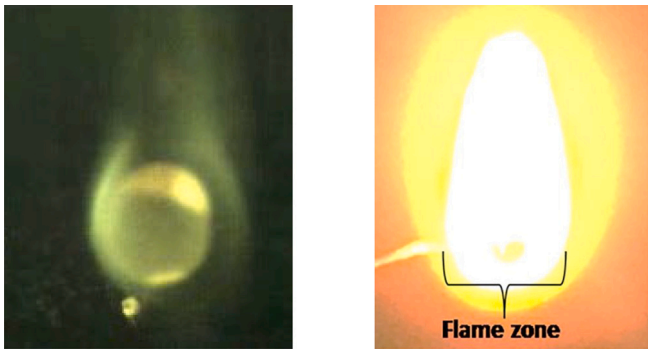


Fig. 1. Experimental images of (left) burning solid aluminum particle [4] and (right) biodiesel droplet [9]. The wake region near the particle/droplet has a different temperature and chemical composition than the surrounding bulk fluid.

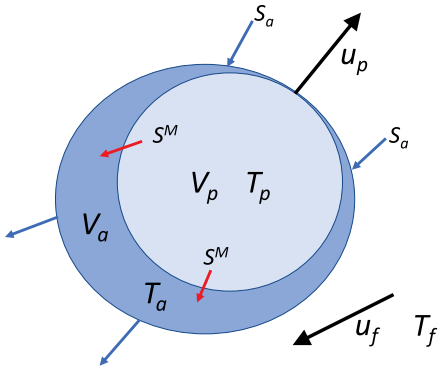


Fig. 2. Schematic of a solid particle with volume V_p surrounded by its added volume V_a, both of which have velocity $\mathbf{u}_p$. The bulk fluid surrounding the particle has velocity $\mathbf{u}_f$. The added volume remains in equilibrium with the bulk fluid due to the mass-transfer term S_a. In this work, we include mass transfer from the particle to the added mass denoted by S^M. The temperatures and densities of all phases can be different.

as the particle phase. However, this will not be the case when mass transfers between the particle and added-mass phases due, for example, to phase change.

In a reacting flow the chemical composition and temperature in the added-mass phase can be very different than in the bulk fluid. For example, rapid fluid-phase reactions may occur essentially in the added-mass phase when one of the limiting reactants is present due mass transfer from the particle phase. Such situations are known to occur for high-temperature gas–particle flows [4,17], spray combustion [9,18], and acid–base reactions in bubbly flows [19]. As shown in Fig. 1, the fluid-phase reacting zone surrounding a particle/droplet can be clearly observed in experiments. From a modeling perspective, the added-mass phase provides a structured model for the fluid phase separating the near-particle and bulk regions of the multiphase flow. This model structure offers the possibility of using more accurate mass/heat-transfer correlations between the particle and added-mass phase, and between the added-mass and bulk-fluid phases. Although our focus in this work is on heat/mass transfer due to phase change in a single-component system, we expect that the proposed model will especially useful for multiphase flows with fast chemical reactions controlled by heat/mass transfer, such as those shown in Fig. 1.

The remainder of this work is organized as follows. In Section 2, we provide a brief overview of the multiphase flow model with an added-mass phase. Further details can be found in [13,15,20,21]. Then, in Section 3, we describe the new components of the model needed to handle heat and mass transfer with phase change between the particle and added-mass phases. In prior work [14,20,22], we developed a numerical algorithm to solve the multiphase flow model. Thus, in Section 4, we briefly review this algorithm and the changes needed to treat mass transfer. To demonstrate the importance of having the added-mass phase in the presence of mass/heat transfer, the model is applied to sublimation/condensation of carbon dioxide in Section 5. Finally, in Section 6, conclusions are drawn concerning the properties of the multiphase model.

2. Overview of multiphase model with added mass

In this work, we extend the definition of the added-mass phase (see Fig. 2 for details) to account for mass transfer (denoted by S^M) between a particle and its added mass with different temperatures T_p and T_a, respectively. As done in our previous work [13], the added-mass phase exchanges mass and energy at rate S_a with the bulk fluid with temperature T_f. However, since mass transfer with phase change involves a latent heat, the specific enthalpy of the added-mass and bulk fluids will be different. Thus, the principal assumption needed to close the model is that the pressures in the bulk-fluid and added-mass phases are identical and, hence, the added-mass and bulk-fluid densities can be different. The bulk-fluid material density is denoted by ρ_f, the added-mass fluid material density is denoted by ρ_a, and the solids material density is ρ_p (which is assumed constant). In the multiphase model, the bulk-fluid and particle-phase velocities are $\mathbf{u}_f$ and $\mathbf{u}_p$, respectively. The added-mass phase has the same velocity as the particle phase. Here, we consider a single-component fluid so that its mass fraction is unity. However, for multi-component cases, chemical-species mass fractions in each phase can be introduced, as well as homogeneous chemical reactions. (See Appendix A and [22] for details.)

In the extended multiphase model, the specific total energies of the bulk-fluid, added-mass, and particle phases are denoted as E_f, E_a and E_p, respectively. The specific total energy is made up of kinetic and internal energies. We denote the former for each phase as K_f and K_p (= K_a), and the latter as e_f, e_a, and e_p. As we are interested in phase change due to sublimation of a solid to a gas (e.g., for cooling electronic components [6]), we introduce the energy of sublimation e_s and include it in the definition of e_p. Hence, the specific total energies are

$$E_f = K_f + e_f, \quad E_a = K_p + e_a, \quad E_p = K_p + e_p. \tag{1}$$

The kinetic energy is further divided into mean and fluctuating components:

$$K_f = \frac{1}{2}u_f^2 + k_f, \quad K_p = \frac{1}{2}u_p^2 + \frac{3}{2}\Theta_p \tag{2}$$

where $u^2 = \mathbf{u}\cdot\mathbf{u}$, k_f is the pseudo-turbulent kinetic energy (PTKE), and Θ_p is the granular temperature. Because the added-mass travels with the individual particles, we find that $K_a = K_p$. However, $E_a \neq E_p$ because the specific internal energy of the solid e_p and fluid e_a are different. The thermodynamic temperatures T_f, T_a, and T_p are found from the internal energies e_f, e_a, and e_p+e_s, respectively (see Table 2).[1]

In order to determine the density ρ_a of the added-mass phase, we assume that its pressure p_a is equal to the bulk-fluid pressure p_f.[2] For example, using the ideal-gas law for a pure gas, the densities are related by $\rho_a T_a = \rho_f T_f$. In prior work, we assumed that $\rho_f = \rho_a$ (which is a good approximation for liquids). Thus, in this work, we must solve an additional transport equation for e_a in order to find T_a. In any case, we will use the same notation for the volume fractions, i.e., α_p for the particle phase, α_a for the added-mass phase, and $\alpha_f^\star$ for the bulk-fluid

[1] The constant heat capacity expressions are given in Table 2. If the exact formulas are used to relate the internal energy to the temperature [22], the latent heat of sublimation arises due to phase change.

[2] In a bubbly flow, ρ_p is for a gas/vapor phase and its value can be found by assuming its pressure is related to the external pressure p_f and surface tension.

Table 1
11-equation model for fluid–particle flow with added mass, PTKE and mass transfer. Gravity $\mathbf{g}$ is included in the model for completeness but is negligible for high-speed flows. The solid material density ρ_p is constant.

Number-density and mass balances:

$$\partial_t N_d + \partial_{\mathbf{x}} \cdot (N_d \mathbf{u}_p) = 0$$
$$\partial_t(\rho_p \alpha_p) + \partial_{\mathbf{x}} \cdot (\rho_p \alpha_p \mathbf{u}_p) = -S^M$$
$$\partial_t(\rho_a \alpha_a) + \partial_{\mathbf{x}} \cdot (\rho_a \alpha_a \mathbf{u}_p) = S_a + S^M$$
$$\partial_t(\rho_f \alpha_f^\star) + \partial_{\mathbf{x}} \cdot (\rho_f \alpha_f^\star \mathbf{u}_f) = -S_a$$

Momentum balances:

$$\partial_t(\rho_f \alpha_f^\star \mathbf{u}_f) + \partial_{\mathbf{x}} \cdot [\rho_f \alpha_f^\star \mathbf{u}_f \otimes \mathbf{u}_f + \hat{p}_f \mathbf{I} + \alpha_p^\star \rho_f (\Theta_p \mathbf{I} + \alpha_f^\star \mathbf{R})] = \alpha_p^\star (\partial_{\mathbf{x}} \hat{p}_f + \mathbf{F}_{pf}) + \frac{\rho_e \alpha_p^\star}{\tau_p} \mathbf{u}_{pf} - \mathbf{S}_{\mathbf{u}} + \rho_f \alpha_f^\star \mathbf{g}$$

$$\partial_t(\rho_e \alpha_p^\star \mathbf{u}_p) + \partial_{\mathbf{x}} \cdot [\rho_e \alpha_p^\star (\mathbf{u}_p \otimes \mathbf{u}_p + \Theta_p \mathbf{I} + P_p \mathbf{I}) + \mathbf{P}_{pfp}] = -\alpha_p^\star (\partial_{\mathbf{x}} \hat{p}_f + \mathbf{F}_{pf}) - \frac{\rho_e \alpha_p^\star}{\tau_p} \mathbf{u}_{pf} + \mathbf{S}_{\mathbf{u}} + \rho_e \alpha_p^\star \mathbf{g}$$

Bulk fluid-phase energy balances (total and PTKE):

$$\partial_t(\rho_f \alpha_f^\star E_f) + \partial_{\mathbf{x}} \cdot [\rho_f \alpha_f^\star \mathbf{u}_f E_f + (\alpha_f^\star \mathbf{u}_f + \alpha_p^\star \mathbf{u}_p)\hat{p}_f + \alpha_p^\star \rho_f (\Theta_p \mathbf{I} + \alpha_f^\star \mathbf{R}) \cdot \mathbf{u}_p] = -\mathbf{P}_{pfp} : (\partial_{\mathbf{x}} \mathbf{u}_p) + \alpha_p^\star \mathbf{u}_p \cdot (\partial_{\mathbf{x}} \hat{p}_f + \mathbf{F}_{pf}) + \frac{\rho_e \alpha_p^\star}{\tau_p} [3a\Theta_p - 2(1-a)k_f + \mathbf{u}_{pf} \cdot \mathbf{u}_p] + S_{af}^E - S_E + \rho_f \alpha_f^\star \mathbf{u}_f \cdot \mathbf{g}$$

$$\partial_t(\rho_f \alpha_f^\star k_f) + \partial_{\mathbf{x}} \cdot (\rho_f \alpha_f^\star \mathbf{u}_f k_f) = -\frac{2}{3} \rho_f \alpha_f^\star k_f \partial_{\mathbf{x}} \cdot \mathbf{u}_f + \frac{\rho_e \alpha_p^\star}{\tau_p} [3a\Theta_p - 2(1-a)k_f + u_{pf}^2 - C_f k_f] - S_k$$

Particle-phase energy balances (kinetic and internal):

$$\partial_t(\rho_e \alpha_p^\star K_p) + \partial_{\mathbf{x}} \cdot [\rho_e \alpha_p^\star (K_p + \Theta_p + P_p)\mathbf{u}_p + \mathbf{P}_{pfp} \cdot \mathbf{u}_p] = \mathbf{P}_{pfp} : (\partial_{\mathbf{x}} \mathbf{u}_p) - \alpha_p^\star \mathbf{u}_p \cdot (\partial_{\mathbf{x}} \hat{p}_f + \mathbf{F}_{pf}) - \frac{\rho_e \alpha_p^\star}{\tau_p} [3a\Theta_p - 2(1-a)k_f + \mathbf{u}_{pf} \cdot \mathbf{u}_p] - H_p + S_K + \rho_e \alpha_p^\star \mathbf{u}_p \cdot \mathbf{g}$$

$$\partial_t(\rho_p \alpha_p e_p) + \partial_{\mathbf{x}} \cdot (\rho_p \alpha_p e_p \mathbf{u}_p) = H_p - S_{pa}^E$$
$$\partial_t(\rho_a \alpha_a e_a) + \partial_{\mathbf{x}} \cdot (\rho_a \alpha_a e_a \mathbf{u}_p) = S_{pa}^E - S_{af}^E + S_e$$

phase. These volume fractions are related by $\alpha_f + \alpha_p = 1$ and $\alpha_f^\star + \alpha_p^\star = 1$. Thus, for example, $\alpha_p^\star = \alpha_p + \alpha_a$ and $\alpha_f = \alpha_f^\star + \alpha_a$.

In the momentum balance for the particle phase, we use the notation $\rho_e \alpha_p^\star = \rho_p \alpha_p + \rho_a \alpha_a$ to represent the mass where the effective density can be expressed in terms of $\alpha_p / \alpha_p^\star$ as

$$\rho_e = \frac{\alpha_p}{\alpha_p^\star}(\rho_p - \rho_a) + \rho_a. \tag{3}$$

In the model with added mass, it is necessary to solve three mass balances [13]. For example, the conserved mass variables can be chosen as $X_1 = \rho_p \alpha_p$, $X_2 = \rho_a \alpha_a$ and $X_3 = \rho_f \alpha_f^\star$. As noted above, the density ratio $\kappa = \rho_a / \rho_f$ is known and ρ_p is constant. Thus, given the conserved variables, the primitive variables can be found in the following order:

$$\alpha_p = \frac{X_1}{\rho_p}, \quad \alpha_f = 1 - \alpha_p, \quad \hat{\kappa} = \frac{X_2}{X_3}, \quad \alpha_a = \frac{\hat{\kappa}}{\kappa + \hat{\kappa}} \alpha_f, \quad \alpha_p^\star = \alpha_p + \alpha_a,$$
$$\alpha_f^\star = 1 - \alpha_p^\star, \quad \rho_f = \frac{X_3}{\alpha_f^\star}, \quad \rho_a = \kappa \rho_f. \tag{4}$$

These formulas reduce to the ones given in our previous work when $\kappa = 1$.

With the definitions given above, we can now define the mixture variables for mass, momentum and total energy for the dispersed, two-phase system:

$$\rho_m = \alpha_p^\star \rho_e + \alpha_f^\star \rho_f = \alpha_p \rho_p + \alpha_a \rho_a + \alpha_f^\star \rho_f,$$
$$\rho_m \mathbf{u}_m = \alpha_p^\star \rho_e \mathbf{u}_p + \alpha_f^\star \rho_f \mathbf{u}_f, \tag{5}$$
$$\rho_m E_m = \alpha_p^\star \rho_e K_p + \alpha_p \rho_p e_p + \alpha_a \rho_a e_a + \alpha_f^\star \rho_f E_f;$$

and the volume-average velocity variables:

$$\mathbf{u}_v = \alpha_p^\star \mathbf{u}_p + \alpha_f^\star \mathbf{u}_f,$$
$$\boldsymbol{\Gamma}_v = \frac{1}{2}[(\partial_{\mathbf{x}} \mathbf{u}_v) + (\partial_{\mathbf{x}} \mathbf{u}_v)^t], \tag{6}$$
$$\mathbf{S}_v = \boldsymbol{\Gamma}_v - \frac{1}{3}(\partial_{\mathbf{x}} \cdot \mathbf{u}_v)\mathbf{I}$$

where $\partial_{\mathbf{x}} \cdot \mathbf{u}_v = 0$ for constant-density flows. The volume-average variables are needed to define unsteady and lift forces [21]. Thus, in addition to the three conserved variables for mass, we need two conserved variables for momentum: $X_4 = \alpha_p^\star \rho_e \mathbf{u}_p$, $X_5 = \alpha_f^\star \rho_f \mathbf{u}_f$; and five conserved variables for energy: $X_6 = \alpha_p^\star \rho_e K_p$, $X_7 = \alpha_f^\star \rho_f E_f$, $X_8 = \alpha_f^\star \rho_f k_f$, $X_9 = \alpha_p \rho_p e_p$, and $X_{10} = \alpha_a \rho_a e_a$. In comparison to the previous 9-equation model, there is one additional variable X_{10} that arises due to the assumption that $e_a \neq e_p$. In Appendix A, we briefly show how the model can be extended to include N chemical species in each phase; however, the non-isothermal case with a single species already contains much of the physics associated with changes in α_a due to heat and mass transfer.

Finally, if we allow for mass transfer between the particle and added-mass phases, then the particle volume will change. In general, the polydisperse model presented in [20] can be used to describe the particle size distribution. In this work, we will use the simplest version of the polydisperse model that accounts only for the mean particle volume V_p (or diameter d_p) at each time and location. Thus, we will need to solve for the particle number density $X_0 = N_d$,[3] which is related to α_p by

$$\alpha_p = V_p N_d = \frac{\pi}{6} d_p^3 N_d \quad \Longrightarrow \quad d_p = \left(\frac{6\alpha_p}{\pi N_d} \right)^{1/3}. \tag{7}$$

Unlike the variable X_1, X_0 has no source term due to mass transfer. The value of the local particle diameter is then found from Eq. (7) (instead of being fixed when α_p has no source term). Note that this simple model does not allow us to describe polydisperse particles accurately because more than one additional size moment is required to account for size segregation (see [15,20] for examples). However, it suffices for modeling the change in size due to mass transfer. The extended multiphase model therefore uses 11 conserved variables: $\{X_0, \ldots, X_{10}\}$ whose transport equations are reviewed next.

3. Multiphase model with mass transfer

A complete derivation of the 9-equation model starting from a kinetic equation for the particle phase can be found elsewhere [15,20,21]. The principal objective of this section is to present the mass, momentum and kinetic-energy exchange terms in the 11-equation model given in Tables 1 and 2, with emphasis on the new terms arising from mass transfer between the particle and added-mass phases.

[3] In the polydisperse model of [20], $N_d = M_0$.

Table 2
Parameters appearing in the 11-equation model for fluid–particle flow in Table 1. The remaining parameters are defined in the main text. Here, the temperatures are found from the internal energies by assuming that the heat capacities are constant. The exact formula for non-constant heat capacity can be found in [22]. Notice that $e_p = C_{p,p}T_p - e_s$ is always positive, which provides a limit on e_s. The expressions for $\mathbf{P}_{pfp}$ and $\mathbf{F}_{pf}$ are derived in [21] and are necessary for the model to be well posed.

$\mathbf{u}_{pf} = \mathbf{u}_p - \mathbf{u}_f$ $\quad \mathbf{R} = \frac{1}{3}u_{pf}^2\mathbf{I} + \frac{2}{5}\mathbf{u}_{pf}\otimes\mathbf{u}_{pf}$ $\quad Re_p = \frac{d_p v_{pf}}{\nu_f}$ $\quad Re_a = \frac{d_a v_{pf}}{\nu_f}$ $\quad v_{pf} = \frac{\alpha_f^\star}{\alpha_f}u_{pf} + (3\Theta_p)^{1/2}$ $\quad \frac{d_a}{d_p} = \left(\frac{\alpha_p^\star}{\alpha_p}\right)^{1/3}$

$d_p = \left(\frac{6\alpha_p}{\pi N_d}\right)^{1/3}$ $\quad \mathbf{P}_{pfp} = \rho_f(\alpha_p^\star)^2\mathbf{R}$ $\quad \mathbf{F}_{pf} = (\Theta_p\mathbf{I} + \alpha_f^\star\mathbf{R})\cdot\partial_{\mathbf{x}}\rho_f - \frac{2}{3}\rho_f(\partial_{\mathbf{x}}\cdot\mathbf{u}_v)\mathbf{u}_{pf} - \frac{4}{5}\rho_f\mathbf{S}_v\cdot\mathbf{u}_{pf}$

$Nu_p = (7 - 10\alpha_f + 5\alpha_f^2)\left(\frac{d_a+d_p}{d_a-d_p}\right)$ $\quad Nu_a = (7 - 10\alpha_f^\star + 5\alpha_f^{\star 2})(1 + 0.7Re_a^{0.2}Pr_f^{1/3}) + (1.33 - 2.4\alpha_f^\star + 1.2\alpha_f^{\star 2})Re_a^{0.7}Pr_f^{1/3}$

$Pr_f = \frac{\rho_f C_{p,f}\nu_f}{\lambda_f}$ $\quad c_m^\star = \frac{1}{2}\min(1 + 2\alpha_p, 2)$ $\quad \tau_a = C_a\tau_p$ $\quad \tau_p = \frac{4\rho_e d_p^2}{3\mu_f C_D Re_p}$ $\quad C_D = \frac{24}{Re_p}(1 + 0.15Re_p^{0.687})\alpha_f^{-2.65}$

$\tau_c = \frac{d_p\sqrt{\pi}}{12\alpha_p g_0\Theta_p^{1/2}}$ $\quad \tau_{fr} = \frac{c_{fr}}{\max(|\partial_{\mathbf{x}}\cdot\mathbf{u}_p|, 1/\tau_c)h_{fr}}$ $\quad g_0 = \frac{1+\alpha_f}{2\alpha_f^3}$ $\quad h_{fr} = \frac{1}{2}\left[1 + \tanh\left(\frac{\alpha_p - \alpha_{max}}{\Delta_{fr}}\right)\right]$

$p_f = (\gamma_f - 1)\rho_f e_f$ $\quad e_f = E_f - \frac{1}{2}u_f^2 - k_f$ $\quad \Theta_p = \frac{2}{3}K_p - \frac{1}{3}u_p^2$ $\quad H_p = \rho_e\alpha_p^\star\left[\frac{1}{\tau_c}(1 - e_c^2) + \frac{1}{\tau_{fr}}\right]\Theta_p$

$\hat{p}_f = p_f + \frac{2}{3}\rho_f k_f$ $\quad P_p = P_c + P_{fr}$ $\quad P_c = 2(1 + e_c)\alpha_p g_0\Theta_p$ $\quad P_{fr} = \frac{p_{fr}\alpha_p g_0 h_{fr}}{\rho_e\alpha_p^\star}$ $\quad T_f = \frac{\gamma_f e_f}{C_{p,f}}$

$T_a = \frac{\gamma_a e_a}{C_{p,a}}$ $\quad T_p = \frac{e_p + e_s}{C_{p,p}}$ $\quad C_f^{-1} = \alpha_p[1 + 1.25\alpha_f^3\exp(-\alpha_p\alpha_f^{1/2}Re_p^{1/2})]$ $\quad \frac{1-a}{1-a_{min}} = \frac{\rho_f}{\rho_p K + \rho_f}$

3.1. Multiphase model for the particle phase

The number-density and mass balances for the particle and added-mass are, respectively,

$$\partial_t N_d + \partial_{\mathbf{x}}\cdot(N_d\mathbf{u}_p) = 0, \tag{8}$$

$$\partial_t(\rho_p\alpha_p) + \partial_{\mathbf{x}}\cdot(\rho_p\alpha_p\mathbf{u}_p) = -S^M, \tag{9}$$

and

$$\partial_t(\rho_a\alpha_a) + \partial_{\mathbf{x}}\cdot(\rho_a\alpha_a\mathbf{u}_p) = S_a + S^M \tag{10}$$

where S_a is the added-mass exchange term, already present in the 9-equation model (but with $\rho_a = \rho_f$) [15]. The new mass-exchange term S^M represent solid mass moved to the gas in the added-mass phase. Because $\rho_f \ll \rho_p$, positive S^M will cause α_a to grow. In response, S_a will be negative to move α_a back to its equilibrium value. The fact that S^M does not appear in the bulk-fluid mass balance is the first major difference with the standard two-fluid model. Of course, if Eq. (10) is added to the bulk-fluid mass balance, then S^M will appear while S_a will not.

The momentum balance for the particle phase, including the added-mass phase, is the same as in the 9-equation model [15]:

$$\partial_t(\rho_e\alpha_p^\star\mathbf{u}_p) + \partial_{\mathbf{x}}\cdot[\rho_e\alpha_p^\star(\mathbf{u}_p\otimes\mathbf{u}_p + \Theta_p\mathbf{I} + P_p\mathbf{I}) + \mathbf{P}_{pfp}] = -\alpha_p^\star(\partial_{\mathbf{x}}\hat{p}_f + \mathbf{F}_{pf}) - \mathbf{D} + \mathbf{S}_{\mathbf{u}}. \tag{11}$$

Notice that there is no term equivalent to S^M is this balance because $\mathbf{u}_p$ is the same for particles and added mass. As in the 9-equation model, $\mathbf{S}_{\mathbf{u}}$ is related to S_a. Likewise, the kinetic-energy balance is the same as in the 9-equation model:

$$\partial_t(\rho_e\alpha_p^\star K_p) + \partial_{\mathbf{x}}\cdot[\rho_e\alpha_p^\star(K_p + \Theta_p + P_p)\mathbf{u}_p + \mathbf{P}_{pfp}\cdot\mathbf{u}_p] = \mathbf{P}_{pfp} : (\partial_{\mathbf{x}}\mathbf{u}_p) - \alpha_p^\star\mathbf{u}_p\cdot(\partial_{\mathbf{x}}\hat{p}_f + \mathbf{F}_{pf}) - D_E - H_p + S_K. \tag{12}$$

Again, mass transfer S^M does not appear in this balance equation because $K_p = K_a$.

To complete the 11-equation model, we require balances for e_p and e_a. These have the same form as in the 9-equation model [15]:

$$\partial_t(\rho_p\alpha_p e_p) + \partial_{\mathbf{x}}\cdot(\rho_p\alpha_p e_p\mathbf{u}_p) = H_p - S_{pa}^E \tag{13}$$

and

$$\partial_t(\rho_a\alpha_a e_a) + \partial_{\mathbf{x}}\cdot(\rho_a\alpha_a e_a\mathbf{u}_p) = S_{pa}^E - S_{af}^E + S_e. \tag{14}$$

As before, H_p is particle heating due to inelastic collisions and S_e is related to S_a. The heat-transfer terms S_{pa}^E (from particles to added mass) and S_{af}^E (from added mass to bulk fluid) will be expressed in terms of heat-transfer coefficients. Recall that e_p includes the energy of sublimation e_s.

3.2. Multiphase model for bulk-fluid phase

The bulk-fluid phase is governed by mass, momentum, total-energy, and PTKE balances that have the same form as in the 9-equation model [21]:

$$\partial_t(\rho_f\alpha_f^\star) + \partial_{\mathbf{x}}\cdot(\rho_f\alpha_f^\star\mathbf{u}_f) = -S_a \tag{15}$$

$$\partial_t(\rho_f\alpha_f^\star\mathbf{u}_f) + \partial_{\mathbf{x}}\cdot[\rho_f\alpha_f^\star\mathbf{u}_f\otimes\mathbf{u}_f + \hat{p}_f\mathbf{I} + \alpha_p^\star\rho_f(\Theta_p\mathbf{I} + \alpha_f^\star\mathbf{R})] = \alpha_p^\star(\partial_{\mathbf{x}}\hat{p}_f + \mathbf{F}_{pf}) + \mathbf{D} - \mathbf{S}_{\mathbf{u}} \tag{16}$$

$$\partial_t(\rho_f\alpha_f^\star E_f) + \partial_{\mathbf{x}}\cdot[\rho_f\alpha_f^\star\mathbf{u}_f E_f + (\alpha_f^\star\mathbf{u}_f + \alpha_p^\star\mathbf{u}_p)\hat{p}_f + \alpha_p^\star\rho_f(\Theta_p\mathbf{I} + \alpha_f^\star\mathbf{R})\cdot\mathbf{u}_p] = -\mathbf{P}_{pfp} : (\partial_{\mathbf{x}}\mathbf{u}_p) + \alpha_p^\star\mathbf{u}_p\cdot(\partial_{\mathbf{x}}\hat{p}_f + \mathbf{F}_{pf}) + D_E + S_{af}^E - S_E \tag{17}$$

$$\partial_t(\rho_f\alpha_f^\star k_f) + \partial_{\mathbf{x}}\cdot(\rho_f\alpha_f^\star\mathbf{u}_f k_f) + \frac{2}{3}\rho_f\alpha_f^\star k_f\partial_{\mathbf{x}}\cdot\mathbf{u}_f = D_{PT} - \rho_e\alpha_p^\star\frac{C_f}{\tau_p}k_f - S_k. \tag{18}$$

The only 'new' term[4] is the energy exchange with the added-mass phase S_{af}^E. The fluid pressure is found from an equation of state. In this work, we will use the ideal-gas law:

$$p_f = (\gamma_f - 1)\rho_f\left(E_f - K_f\right) = (\gamma_f - 1)\rho_f e_f \tag{19}$$

where γ_f is the heat-capacity ratio. Recall that an important modeling assumption is that $p_a = p_f$ such that

$$(\gamma_f - 1)\rho_f e_f = (\gamma_a - 1)\rho_a e_a \quad\Longrightarrow\quad \kappa = \frac{(\gamma_f - 1)e_f}{(\gamma_a - 1)e_a}, \tag{20}$$

which fixes the value of ρ_a given ρ_f. In this work, we assume $\gamma_a = \gamma_f$ so that $\kappa = e_f/e_a$.

3.3. Models for exchange terms

The terms $\mathbf{D}$ and D_{PT} represent drag exchange:

$$\mathbf{D} = \frac{\rho_e\alpha_p^\star}{\tau_p}\mathbf{u}_{pf}, \quad D_{PT} = \frac{\rho_e\alpha_p^\star}{\tau_p}[u_{pf}^2 + 3a\Theta_p - 2(1 - a)k_f]. \tag{21}$$

Physically, $D_{PT} \geq 0$ represents kinetic energy lost from the particle phase to produce PTKE in the fluid phase. These models are the same as in the 9-equation model [15]. For completeness, the 11-equation model and the corresponding closures are provided in Tables 1 and 2. For the

[4] The term S_k was not included in our previous publications, but is needed to keep k_f constant in homogeneous cases.

most part, these models are the same as in our previous work without mass transfer.

The mass-exchange term $S^M = \dot{M}$ corresponds to phase change from particles to fluid $\dot{M} > 0$ or fluid to particles $\dot{M} < 0$ where the fluid is in the added-mass phase. The only necessary conditions are that $\dot{M} \leq 0$ when $\alpha_p = 0$ and $\dot{M} \geq 0$ when $\alpha_a = 0$. In this work, we will consider a simple rate expression that satisfies these constraints. However, more generally, one would use a physics-based model for evaporation/condensation or surface reactions to determine $\dot{M}$ [22]. Unlike in [22], there is no momentum transfer corresponding to $\dot{M}$ because the momentum balance includes the particle and the added mass, i.e., $\rho_e \alpha_p^\star$, for which S^M has no contribution.

Parameter values used in the test case are provided in Table 3. In order to model CO_2 sublimation, we will use the following expression for $\dot{M}$ [6]:

$$\dot{M} = \begin{cases} K_s \rho_p \alpha_p & \text{if } K_s > 0 \\ K_s \rho_a \alpha_a & \text{if } K_s < 0 \end{cases} \tag{22}$$

and

$$K_s = K_0 \frac{6 \lambda_a N u_p}{\rho_p e_s d_p^2} (T_p - T_s) \tag{23}$$

where $T_s = 195$ K is the sublimation temperature of CO_2 and e_s is the specific latent energy. In words, if $T_p > T_s$ the solid will sublimate to create vapor with $K_s > 0$. Otherwise, the vapor will condense ($K_s < 0$) to form the solid. The mass-transfer-rate constant K_s is specified by assuming that heat transfer between the particle and added-mass phases controls the rate of sublimation. For simplicity, in Eq. (22) we have taken $\dot{M}$ to be proportional to the mass of the phase from which the transfer originate. This ensures that $\dot{M} \to 0$ when the donor phase is depleted. In this work, K_0 is a constant used to test different sublimation rates. However, in real systems, K_s is a molecular-scale process so that K_0 will normally be large enough to keep the particle temperature T_p very near T_s, while the added-mass temperature T_a varies due to the latent heat.

The terms S_{pa}^E and S_{af}^E represent energy transfer from the particles to their added masses and from the added masses to the bulk fluid, respectively. For the former, since there is no relative velocity between the two phases and the added volume has finite size, we approximate the Nusselt number Nu_p as given in Table 2. The resulting expression for S_{pa}^E, including the mass-transfer contribution, is

$$S_{pa}^E = \frac{6 \alpha_p \lambda_a N u_p}{d_p^2} (T_p - T_a) + \max(\dot{M}, 0)\, e_p + \min(\dot{M}, 0)\, e_a \tag{24}$$

where λ_a is the conductivity in the added-mass fluid. Notice that because e_s is included in the definition of T_p in Table 2, the energy of sublimation will be accounted for in the mass-transfer terms involving $\dot{M}$. In other words, when $\dot{M} > 0$, the added-mass phase will cool down because at fixed temperature $e_p < e_a$, while for $\dot{M} < 0$ the particle phase will heat up. Likewise, for the second term there is no mass transfer and so we use

$$S_{af}^E = \frac{6 \alpha_a \lambda_f N u_a}{d_a^2} (T_a - T_f) \tag{25}$$

where the diameter of the added volume is estimated by $d_a = d_p (\alpha_p^\star / \alpha_p)^{1/3}$ and Nu_a is the Nusselt number depending on the fluid Prandtl number, the slip Reynolds number, and the fluid-phase volume fraction (see Table 2 for definition).

Finally, the added-volume relaxation rate is defined by[5]

$$r_a = \frac{1}{\tau_a} (c_m^\star \alpha_f \alpha_p - \alpha_a) \tag{26}$$

where r_a is the relaxation rate with timescale τ_a that is smaller than but proportional to the drag timescale τ_p. Recall that this relaxation process is primarily a fluid-dynamical process as opposed to a phase change. Thus, the competition between the rate of phase change $\dot{M}$ and relaxation S_a will determine the value of α_a. Given r_a, the remaining exchange terms are

$$\begin{aligned} S_a &= \max(r_a, 0)\, \rho_f + \min(r_a, 0)\, \rho_a, \\ \mathbf{S_u} &= \max(r_a, 0)\, \rho_f \mathbf{u}_f + \min(r_a, 0)\, \rho_a \mathbf{u}_p, \\ S_K &= \max(r_a, 0)\, \rho_f K_f + \min(r_a, 0)\, \rho_a K_p, \\ S_k &= \max(r_a, 0)\, \rho_f k_f + \min(r_a, 0)\, \frac{3}{2} \rho_a \Theta_p, \\ S_e &= \max(r_a, 0)\, \rho_f e_f + \min(r_a, 0)\, \rho_a e_a, \end{aligned} \tag{27}$$

and $S_E = S_K + S_e$. Except for S_k, these expressions are the same as in the 9-equation model when $\rho_a = \rho_f$ [15].

4. Solution algorithm

The numerical solution of the 11-equation model is obtained by extending the solver presented in [14] to include phase change and heat transfer in the added-mass phase. The additional equations of the proposed model do not modify the eigenvalue structure of the 9-equation model for which the numerical scheme has been initially developed as N_d and e_a are passively transported with speed $\mathbf{u}_p$. Moreover, the source terms to account for phase change can be easily incorporated in the solver by using an operator splitting for temporal integration.

The model equations summarized in Table 1 can be written as

$$\partial_t \mathbf{X} + \partial_{\mathbf{x}} \cdot \boldsymbol{f}(\mathbf{X}) = \boldsymbol{h}(\mathbf{X}) + \boldsymbol{s}(\mathbf{X}) \tag{28}$$

with the vector of conserved variables $\mathbf{X}$, the conservative fluxes $\boldsymbol{f}(\mathbf{X})$, the non-conservative terms of the hyperbolic part of the system $\boldsymbol{h}(\mathbf{X})$, and the local source terms $\boldsymbol{s}(\mathbf{X})$. The integration of Eq. (28) is sequentially performed using a Strang splitting between the hyperbolic operator $\mathcal{H}_{\Delta t}$ and the source-term operator $\mathcal{S}_{\Delta t}$:

$$\mathbf{X}^{n+1} = \mathcal{H}_{\Delta t/2} \mathcal{S}_{\Delta t} \mathcal{H}_{\Delta t/2} \mathbf{X}^n. \tag{29}$$

The hyperbolic operator consists of conservative fluxes and non-conservative terms, while the source-term operator contains the remaining local exchanges between phases or between mean and fluctuating energy.

4.1. Hyperbolic step

As presented in [14], the hyperbolic step $\mathcal{H}_{\Delta t}$ is obtained by applying the Godunov scheme to Eq. (28) without source terms [23]:

$$\mathbf{X}_i^{n+1} = \mathbf{X}_i^n - \frac{\Delta t}{\Delta x} \left(\boldsymbol{f}_{i+\frac{1}{2}} - \boldsymbol{f}_{i-\frac{1}{2}} \right) + \Delta t\, \boldsymbol{h}_i \tag{30}$$

where n and $n+1$ superscripts stand for quantities at time t and $t + \Delta t$, respectively. The subscripts i, $i-\frac{1}{2}$ and $i+\frac{1}{2}$ are the volume average, left face, and right face, respectively, reconstruction of a quantity defined in the ith computational cell with Δx the cell size. The face reconstructions are obtained from WENO5 interpolations [24] with the slope limiter from [25].

As proposed by [22], the system of equations is split into a fluid sub-system solved by the HLLC scheme [26], and a particle subsystem solved by the AUSM$^+$up scheme [27]. The fluid system is solved by freezing the particle-phase volume fraction during a time step and the fluid-phase volume fraction is reconstructed at the face from $\alpha_p^\star = 1 - \alpha_f^\star$ to ensure consistency between the amount of particles seen from both phases. The reconstruction of fluid variables $\mathbf{Y}_f = (\rho_f, u_f, \hat{p}_f, E_f, k_f, P_{pfp})$ is based on the HLLC scheme [22]. The fluid fluxes are finally assembled using the face reconstruction of $\alpha_f^\star$ and $\mathbf{Y}_f$. As explained in [14], the pfp pressure P_{pfp} is also reconstructed from the HLLC scheme resulting in an upwinding.

[5] The usual added-mass constant for two-fluid models is $C_M = c_m^\star \alpha_f$, and is close to 1/2 when $\alpha_p < 0.5$.

Table 3
Model constants for CO_2 gas and particles at $T_s = 195$ K.

$d_0 = 30$ μm	$e_c = 0.9$	$a_{min} = 0.5$	$K = 0.06$	$C_a = 0.001$
$c_{fr} = 0.01$	$\alpha_{max} = 0.63$	$\Delta_{fr} = 0.01$	$p_{fr} = 533,333$ kg/m/s^2	
$\nu_f = 0.847 \times 10^{-5}$ m^2/s	$\gamma_f = \gamma_a = 1.37$	$\lambda_f = \lambda_a = 0.0166$ kg m/s^3/K	$M_w = 44$ kg/kmol	
$e_s = \frac{1}{2} C_{p,p} T_s$ m^2/s^2	$C_{p,f} = C_{p,a} = 849$ m^2/s^2/K	$C_{p,p} = 619$ m^2/s^2/K	$\rho_p = 1562$ kg/m^3	

The extension of the algorithm used in [14] to solve the particle system in the presence of phase chance and heat exchange in the added mass is now detailed. The AUSM$^+$up scheme applied to the particle subsystem leads to the particle flux vector:

$$\boldsymbol{f}_p = \dot{\mathbf{m}}_p \boldsymbol{\Psi}_p + \boldsymbol{\Pi}_p \tag{31}$$

with the mass flux $\dot{\mathbf{m}}_p = (\dot{m}_0, \dot{m}_p, \dot{m}_a, \dot{m}_p^\star, \dot{m}_p^\star, \dot{m}_p, \dot{m}_a)$, the quantity vector $\boldsymbol{\Psi}_p = (1, 1, 1, u_p, K_p, e_p, e_a)$, and the pressure flux $\boldsymbol{\Pi}_p = (0, 0, 0, p_p, p_p u_p, 0, 0)$. As particles, added mass, and particle number density are transported at the same speed u_p, a mass flux vector such that

$$\dot{m}_0 = N_d u_p, \quad \dot{m}_p = \rho_p \alpha_p u_p, \quad \dot{m}_a = \rho_a \alpha_a u_p, \quad \dot{m}_p^\star = \rho_e \alpha_p^\star u_p, \tag{32}$$

is used instead of the single mass flux used in the original version [27]. Moreover, the kinetic energy K_p is solved instead of pseudo-thermal energy introduced in [22], and the unconventional term $p_p u_p$ is added in $\boldsymbol{\Pi}_p$ [14]. The computation of the face compaction wave speed $a_p^{1/2}$, the face Mach number $\mathcal{M}_p^{1/2}$, and the pressure flux $\boldsymbol{\Pi}_p$ are detailed in [14] and follow the methodology in [28]. The quantity vector is simply upwinded with face velocity $u_p^{1/2} = a_p^{1/2} \mathcal{M}_p^{1/2}$:

$$\boldsymbol{\Psi}_p = \begin{cases} \boldsymbol{\Psi}_p^L, & \text{if } u_p^{1/2} \le 0 \\ \boldsymbol{\Psi}_p^R, & \text{otherwise} \end{cases} \tag{33}$$

with $\boldsymbol{\Psi}_p^L$ and $\boldsymbol{\Psi}_p^R$ the left and right reconstructions of $\boldsymbol{\Psi}_p$ at a given face. The fluxes of particle number density, particle mass, and added mass are computed from the face velocity $u_p^{1/2}$ as

$$\dot{m}_0 = u_p^{1/2} \begin{cases} N_d^L, & \text{if } u_p^{1/2} \le 0 \\ N_d^R, & \text{otherwise,} \end{cases} \quad \dot{m}_p = u_p^{1/2} \begin{cases} \rho_p \alpha_p^L, & \text{if } u_p^{1/2} \le 0 \\ \rho_p \alpha_p^R, & \text{otherwise,} \end{cases}$$
$$\dot{m}_a = u_p^{1/2} \begin{cases} \rho_a^L \alpha_a^L, & \text{if } u_p^{1/2} \le 0 \\ \rho_a^R \alpha_a^R, & \text{otherwise,} \end{cases} \tag{34}$$

and the effective mass flux is obtained from

$$\dot{m}_p^\star = \dot{m}_p + \dot{m}_a. \tag{35}$$

These definitions ensure that particle mass, added mass, and number density are discretely consistent and transported at the same speed $u_p^{1/2}$. In addition, the reconstruction of added-mass density needs to account for the new equilibrium $p_a = p_f$, in place of $\rho_a = \rho_f$ used in previous works [13,14]. A consistent reconstruction of ρ_a needs to verify Eq. (20) at the face, which for the ideal-gas law leads to

$$\rho_a^K = \frac{p_f^{HLLC}}{(\gamma_a - 1) e_a^K} \tag{36}$$

where p_f^{HLLC} is the face pressure obtained from the HLLC scheme and superscript K stands for left (L) or right (R) face reconstructions. Finally, the face reconstructions of fluid and particle quantities are used to compute $\boldsymbol{h}(\mathbf{Y})$. The pressure non-disturbing condition [22] is retrieved by combining the pfp-pressure terms of the kinetic energy of the particle phase Eq. (12) as in [14]:

$$P_{pfp} \partial_x u_p - \partial_x (u_p P_{pfp}) = -u_p \partial_x P_{pfp}. \tag{37}$$

In order to remain stable with WENO5 reconstructions, a RK3-SSP scheme [29] is employed that also preserves realizability [30]. The overall time step is based on the maximum particle and fluid eigenvalues. The eigenvalues are more complicated than for the standard Baer–Nunziato equations and are found numerically using a subroutine for the Jacobian matrix that was generated symbolically. Hyperbolicity of a similar system has been demonstrated in [13], which is applicable for the system presented in this paper.

4.2. Local source terms

The remaining part of the system is the local source term $\boldsymbol{s}(\mathbf{Y})$. The operator consists of successive analytical solutions of linear ODEs broken down for each mechanism with a different relaxation time. For each term, a separate semi-analytical solution is derived and used to update the conserved variables in each phase. The semi-analytical solutions of drag $\mathcal{S}^D$, heat transfer $\mathcal{S}^{HT}$, and particle collision–friction $\mathcal{S}^{CF}$ can be found in [14]. The numerical algorithms used for added mass $\mathcal{S}^{AM}$ and phase change $\mathcal{S}^{PC}$ are detailed in Appendix B. The algorithm for added-mass exchange is slightly different than the one used in our prior work [14,20], and is more robust in the presence of mass transfer. The local source term operator is then written as

$$\mathcal{S}_{\Delta t} = \mathcal{S}_{\Delta t}^{CF} \mathcal{S}_{\Delta t}^{D} \mathcal{S}_{\Delta t}^{HT} \mathcal{S}_{\Delta t}^{PC} \mathcal{S}_{\Delta t}^{AM}. \tag{38}$$

A first advantage of this approach is to impose strict conservation of quantities during the integration time. Also, it alleviates the time constraint, which can be very restrictive in the case of strong shocks interacting with a dispersed phase or rapid phase change.

5. Numerical examples

We will use two numerical examples to illustrate the effect of mass transfer due to sublimation [31]. The model parameters are given in Table 3 where d_0 is the initial particle diameter. In the first example, we assume that the system is spatially homogeneous with zero velocities. The particle phase is initialized with properties at 195 K (i.e., the sublimation temperature), while the fluid phases are set at 300 K. The system is then allowed to evolve in time to thermodynamic equilibrium. In the second example, a Riemann shock-tube problem is simulated with gas on the left side with a pressure jump, and gas and particles on the right side. The initial pressure difference between the two sides generates a shock wave moving into the particles, which causes them to sublimate due to the temperature difference. Depending on the initial particle volume fraction, the dynamics of this system can be quite complex.

5.1. Homogeneous cases

For homogeneous conditions, the adiabatic system evolves in time towards thermal equilibrium. If the initial particle volume fraction is below a certain limit (≈0.1), then the particle phase will disappear at a finite time. Otherwise, the equilibrium system will have two phases. For the latter, we set $\alpha_p = 0.4$ at time zero, and for the former we use $\alpha_p = 0.04$. In all cases, we set $C_a = 0.001$ in the model for τ_a and test different values of K_0 between 0 and 1000. ($K_0 = 0$ corresponds to the previous model with no mass transfer.) As the mean velocities of the two phases are null, the particle Reynolds numbers are also null. Thus, the Nusselt number Nu_a depends only on $\alpha_f^\star$. In general, we shall see that the heat transfer between the bulk and added-mass fluid $\mathcal{S}_{af}^E$ controls the sublimation rate. This can be seen from the temperature plots in Figs. 3 and 4. For the volume fractions, we include the quantity $\alpha_a/(\alpha_p \alpha_f)$, which represents the local added-mass factor c_m.[6]

For the higher initial volume fraction in Fig. 3, the final temperature for all phases in 195 K and the final particle-phase volume fraction

[6] Recall that the added-mass constant is $C_m = c_m \alpha_f$.

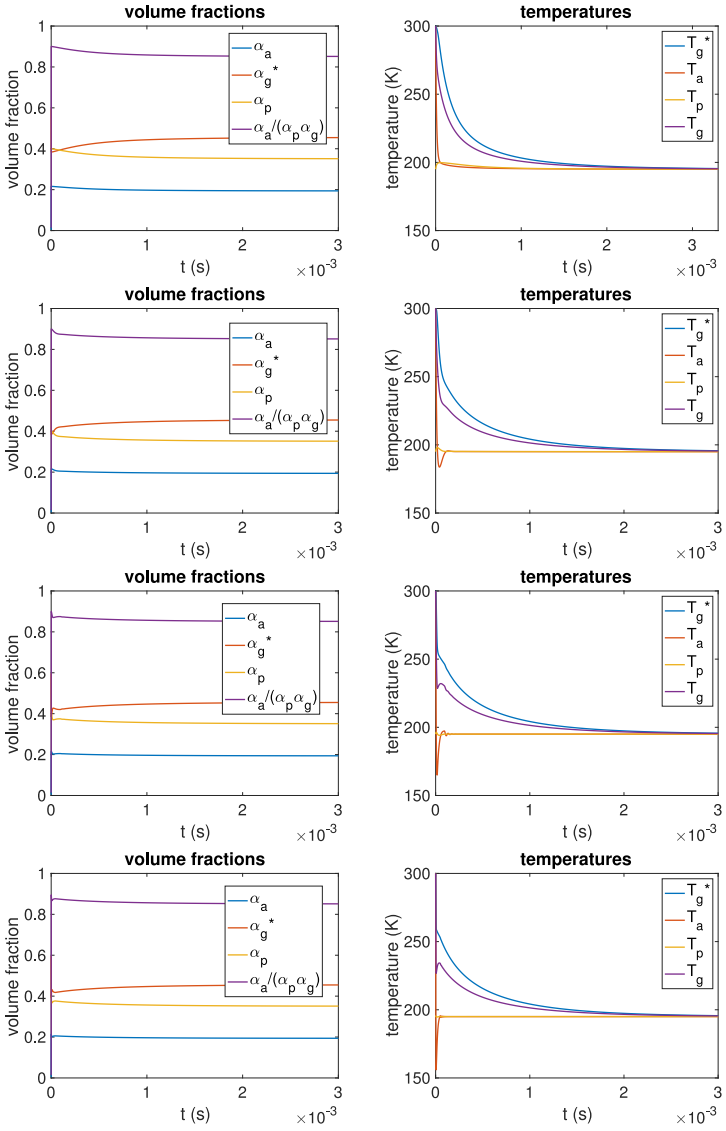


Fig. 3. Volume fraction (left) and temperature (right) versus time for homogeneous case with $\alpha_p(0) = 0.4$. Top to bottom: $K_0 = 1$, 10, 100, 1000. The gas temperature T_g is defined using the mass average of T_a and $T_g^\star$. The effect of K_0 is very small and occurs near $t = 0$ where $T_a < T_s$ is observed for larger K_0.

is the same for all K_0 as expected. At short times, the results depend on K_0. For example, with $K_0 = 1$ the particle-phase temperature T_p is greater than 200 K due to heat transfer from the fluid phase. Physically, one would expect $T_p \approx T_s$ when the sublimation rate is controlled by heat transfer. By increasing K_0, this super heating decreases such that T_p is very near 195 K for the highest K_0. The added-mass temperature T_a is also dependent on K_0 and is super cooled for large K_0 due to the latent heat. In contrast, from Fig. 4 we can note that the process is slower due to the lower particle volume fraction. Moreover, α_p reaches zero at approximately $t = 0.175$ s and the final temperature of the fluid phase is greater than 195 K as expected. Overall, including added mass in the two-fluid model allows for a large temperature difference between the added-mass fluid T_a and the bulk fluid $T_g^\star$. This difference could be important in reacting flows with temperature-dependent reaction rates.

5.2. Shock-tube problem

For the shock tube, the initial fields after the first time step ($t \approx 0$) are shown in Fig. 5. The gas densities on the left and right sides of the domain are 184.96 kg/m^3 and 1.8496 kg/m^3, respectively. Outflow boundary conditions are employed at $x = 0$ and 0.06 m, and a uniform grid with 800 cells ($\Delta x = 7.5 \times 10^{-5}$) is used with CFL number equal to 0.5. The initial velocities and granular temperature are null, the initial particle diameter is set to 30 μm, and the thermodynamic parameters are given in Table 3. Note that the particle phase begins at $x = 0.03$ m, while the shock begins at $x = 0.024$ m. This allows the shock to fully develop before hitting the particles. As expected from the pressure jump, the initial velocity jump in the gas phase at the shock is very large. There is also a strong jump in $\alpha_a/(\alpha_p\alpha_g)$ away from its initial condition of 0.5 that the numerical algorithm handles without difficulty. Finally, due to heat transfer, T_a relaxes towards T_p rather quickly.

In Fig. 6, we present the fields for the base case without sublimation ($K_0 = 0$) at time $t = 0.1$ ms. From the pressure, velocity and temperature fields for the gas phase, one observes the reflected shock near $x = 0.02$ m, whereas the transmitted shock is near $x = 0.047$ m. Ahead of the transmitted shock, the fields remain at their initial conditions, while behind it the particles have been pushed up to close-packed conditions so that P_{fr} sets the particle-phase pressure. Without sublimation, the gas and particles quickly reach the same temperature except at the edge of the transmitted shock where the gas temperature is near 300 K. Note that with the volume fractions, we include the quantity $\alpha_a/(\alpha_p\alpha_f)$,

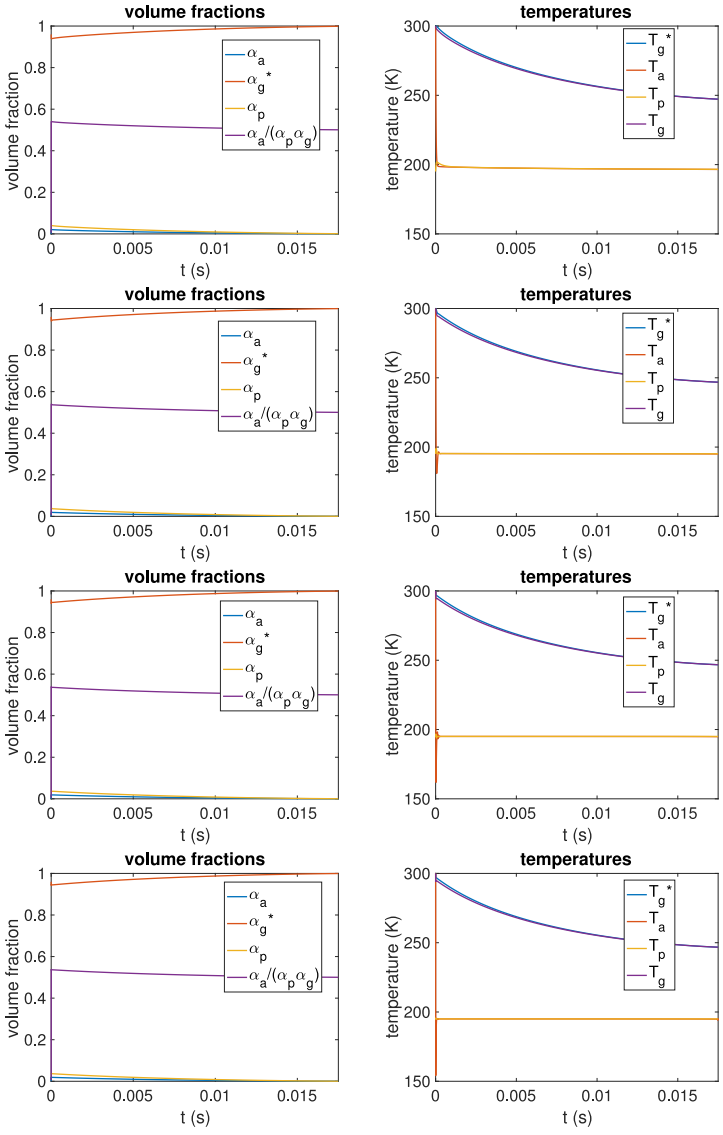


Fig. 4. Volume fraction (left) and temperature (right) versus time for homogeneous case with $\alpha_p(0) = 0.04$. Top to bottom: $K_0 = 1$, 10, 100, 1000. The gas temperature T_g is defined using the mass average of T_a and $T_g^\star$. As the particles become smaller due to sublimation, rapid heat transfer (S_{pa}^E) results in $T_p \approx T_a$. The effect of K_0 is very small and occurs near $t = 0$ where $T_a < T_s$ is observed for larger K_0.

which represents the local added-mass factor c_m. Except at the shock locations, we observe that $c_m = c_m^\star$ as expected. Overall, the base case without sublimation exhibits essentially the same behavior as seen in our prior work [22].

The effect of sublimation is shown in Figs. 7 and 8, which can be compared to the base case in Fig. 6. In the subfigures, we vary K_0 with the values (1,10,100,1000) to increase K_s in Eq. (22). The plots of K_s are shown in Fig. 9, along with the rate of the added-mass source r_a for comparison of their relative magnitudes. Recall that when $K_s > 0$ the mass transfer is from the solid to the gas phase, and $K_s < 0$ corresponds to the gas phase condensing to form the solid phase. With $K_0 = 1$, only sublimation occurs since the particle temperature is always above 195 K behind the transmitted shock. However, already for $K_0 = 10$ we observe a region behind the transmitted shock where the particle temperature is below 195 K so that condensation occurs. For all values of K_0, the particle phase is super-heated at the transmitted shock (i.e., $T_p > 195$ K), which rapidly cools the gas phase below 195 K behind the shock. The higher the value of K_0, the stronger the super-cooling of the gas phase. In fact, it is interesting to note that T_g super-cools more than T_a, which is a result of the time delay for heat transfer between particles, added mass, and the bulk fluid. In any case, the condensation for larger K_0 regenerates the particle phase behind the shock. Indeed, with $K_0 = 1000$ the particles become close-packed and generate a frictional pressure P_{fr} near $x = 0.037$ m.

From Fig. 9 we see that for $K_0 = 1$ and 10, the sublimation rate and r_a have the same order of magnitude. For these cases, the speed of the transmitted shock is faster but not very different from the base case. However, the cases with $K_0 \geq 100$ are dominated by sublimation/condensation and, hence, changing the value of C_a (not shown) has negligible effect. Also, it is interesting to observe how sublimation affects the gas-phase pressure p_g and the velocities. In general, transferring mass from the solid to the gas phase will increase the gas-phase pressure due to the material-density difference (and vice versa for condensation). This is clearly observed for $K_0 = 1000$ and generates a negative velocity, pushing particles backwards relative to the shock in the condensation zone. Indeed, in this case the close-packed particles may eventually move backwards.

The dynamics of the added-mass transfer can best be seen with the volume-fraction distribution at $K_0 = 100$. At the transmitted shock where the sublimation rate is largest, we can observe a spike in $\alpha_a/(\alpha_p\alpha_f)$, which corresponds to α_a being larger than its equilibrium

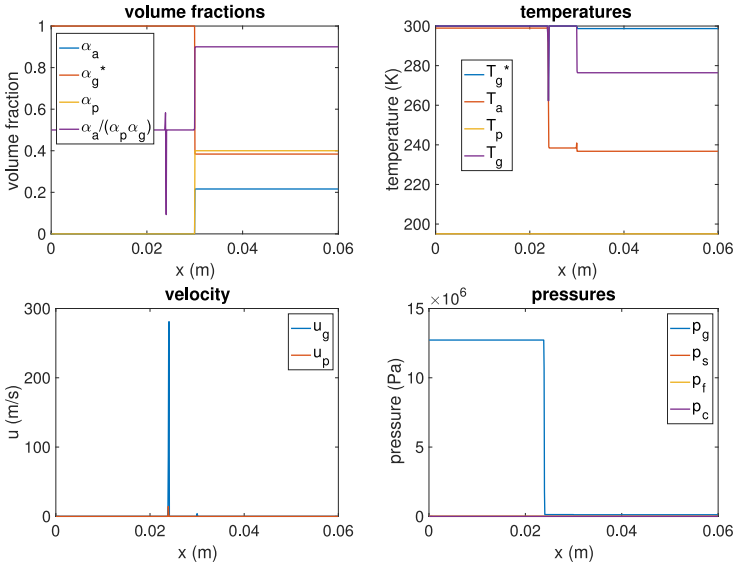


Fig. 5. Fields at first time step $\Delta t \ll 1$, which are very close to the initial conditions. Top left: volume fractions. Top right: temperatures. Bottom left: velocities. Bottom right: pressures. Notice that on the right-hand side of the domain where particles are present the added-mass temperature T_a is changing rapidly towards T_p.

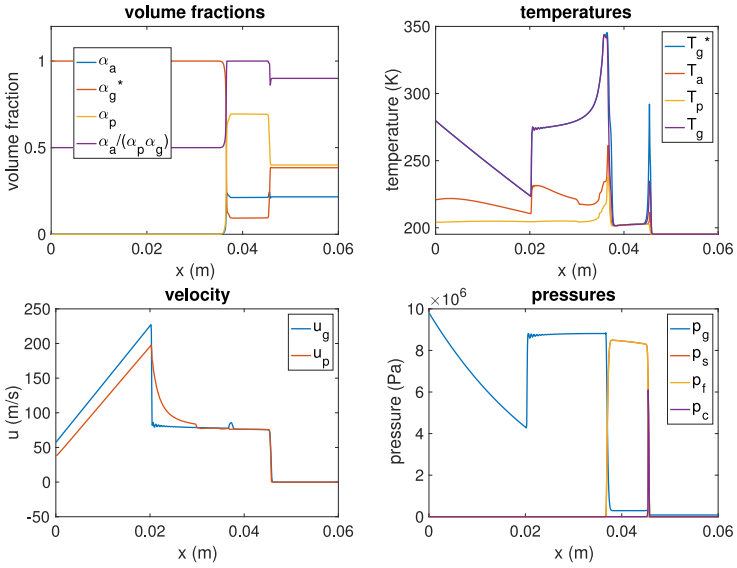


Fig. 6. Base-case distributions for volume fraction (top left), temperature (top right), velocity (bottom left), and pressure (bottom right) with $K_0 = 0$, $C_a = 0.001$ at $t = 0.1$ ms. Notice that around $x = 0.04$ m the particle phase is close packed with large P_{fr}.

value. This is due to mass transfer from the particle to the added-mass phase. Note that at the same location, $\alpha_g^\star$ spikes downward for the same reason. At the same location, the added-mass temperature T_a drops quickly due to the internal energy difference between e_p and e_a. Subsequently, heat transfer pulls down $T_g^\star$ well below 195 K. Heat transfer with the particles then lowers T_p until condensation begins. Note that in this zone $T_p > T_a > T_g^\star$. Likewise, the particle-phase volume fraction α_p has a minimum at the crossover between sublimation and condensation. At this minimum, the particle diameter (see Fig. 10) is an order of magnitude smaller than the initial particle size d_0, thereby significantly enhancing the size-dependent exchange rates.

Position–time plots for the fields α_p, d_p, p_g and u_g are presented in Fig. 10 for the four values of K_0. The x–t plots for $K_0 = 1$ show that d_p, and hence the ratio α_p/N_d, is nearly constant due to the low rate of sublimation. For this case, the shock wave causes α_p to compact up to close-packed conditions, and the highest gas pressure p_g is behind the particles. The x–t plots for $K_0 = 10$ are similar, but now d_p varies across the close-packed region, as does the gas pressure. The latter is due to the exchange of mass from the particles to the gas. This trend continues for $K_0 = 100$, but now the range of d_p values is much larger due to sublimation at the transmitted shock location and condensation behind it. Notice also that the large p_g in the region of rapid sublimation increases u_g substantially in front of the shock. Finally, for $K_0 = 1000$ the strong condensation behind the shock creates a close-packed region at the trailing edge of the particle curtain. Indeed, α_p drops to nearly zero just behind the shock due to rapid sublimation, but then grows again due to rapid condensation. These phenomena have a strong effect on p_g and u_g, whose contours are much more complex than seen with $K_0 = 1$ or 10. In fact, u_g becomes negative due to the high gas pressure in the sublimation region.

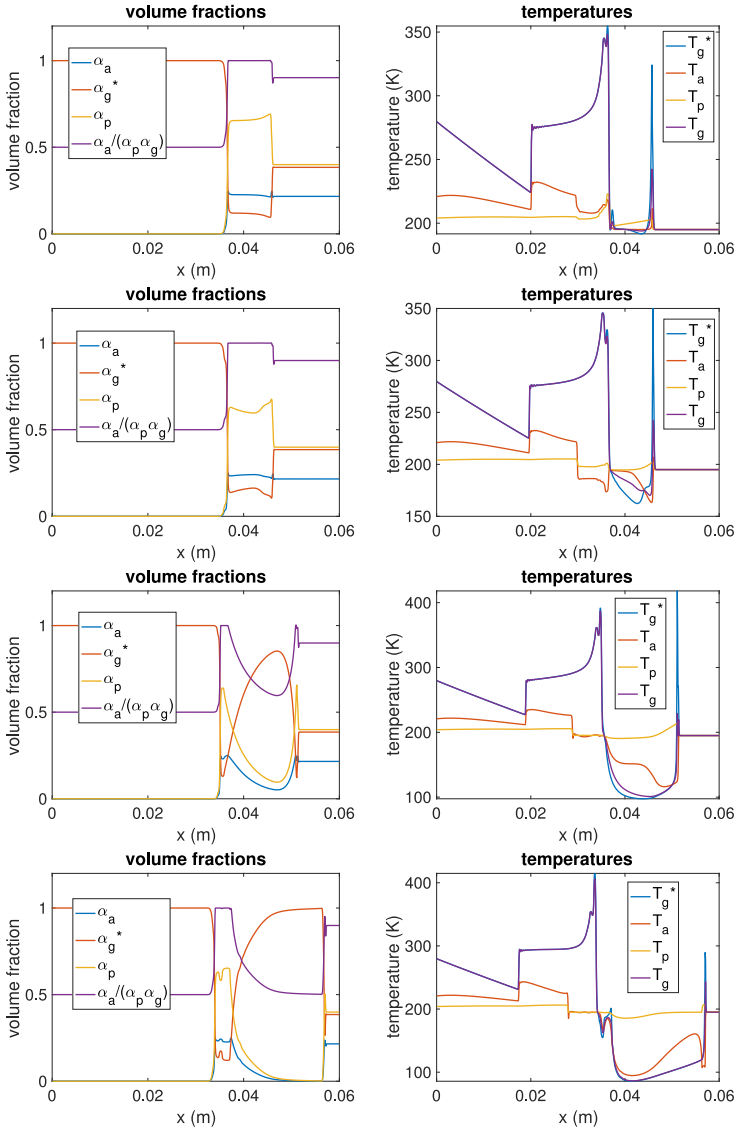


Fig. 7. Volume-fraction (left) and temperature (right) distributions with $C_a = 0.001$ at $t = 0.1$ ms. Top to bottom: $K_0 = 1$, 10, 100, 1000.

In summary, for the case of a shock impinging on a thick particle curtain with mass transfer due to sublimation and condensation, the observed behavior depends strongly on the sublimation rate. When the rate is small, the particle temperature T_p can be significantly larger than T_s, which is unlikely to be physically correct for real systems. Using higher sublimation rates lowers the super-heating of the particles, and changes significantly the flow behavior relative to the base case. At the highest rate tested, sublimation occurs at a sharp interface, behind which condensation is observed. At the sharp interfaces, α_p (and d_p) drops to very small values, before increasing again due to condensation. Remarkably, the gas and particle velocities can be locally negative, and the growing particles become close packed with nonzero frictional pressure. This behavior is mainly due to the large gas-phase pressure generated by rapid sublimation.

Simulations with smaller initial particle volume fraction ($\alpha_p = 0.04$) have also been performed. For this case, the energy in the gas phase is always much higher than in the particle phase, and thus the gas-phase temperature remains much higher than that of the particles. The dynamics then remain close to those seen without sublimation. For example, the large increase in gas-phase pressure is not observed and the sublimation process is controlled by heat transfer from the bulk fluid to the added-mass phase. Eventually, in a very long domain the energy transfer from the gas to the particles, combined with sublimation, will weaken the shock until it comes into equilibrium with the particle phase (as seen in the homogeneous case).

6. Discussion and conclusions

The principal objective of this work was to extend the added-mass model to allow for the added-mass fluid to have a different temperature than the particle phase. In order to achieve this objective, we allowed for the added-mass fluid density to be different than in the bulk fluid by assuming that the fluid pressure is the same in both phases. This change led to a small modification in the algorithms to compute the primitive variables from the conserved variables (see Eq. (4)) and the added-mass transfer (see Appendix B.2). Otherwise, as shown in the shock-tube example, the overall solution algorithm for solving the 11-equation model is robust even for extreme conditions with rapid mass transfer between the solid and gas phases.

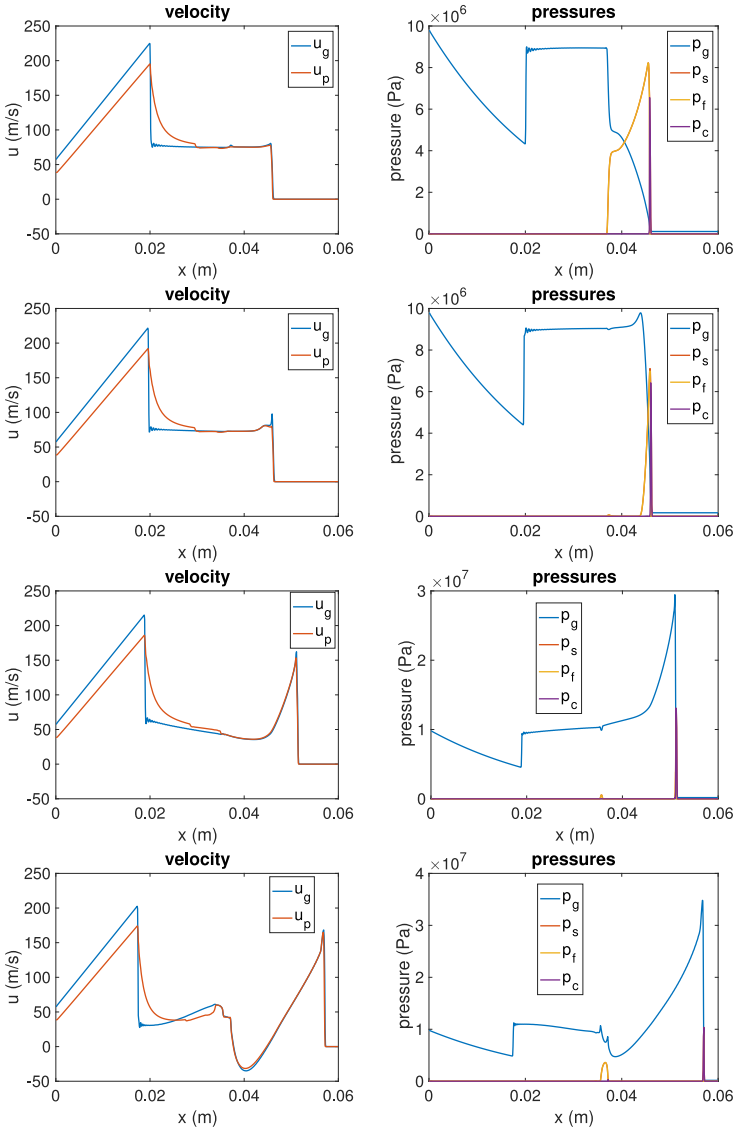


Fig. 8. Volume fraction (left) and pressure (right) distributions with $C_a = 0.001$ at $t = 0.1$ ms. Top to bottom: $K_0 = 1$, 10, 100, 1000. Notice that for $K_0 \geq 100$ the peak gas pressure p_g is significant higher, leading to negative u_g and nonzero P_{fr} for $K_0 = 1000$.

Extending the model to allow for different phase temperatures is the first step towards a multi-component reacting-flow model with added mass. The next step (see Appendix A) will be to add transport equations for the species mass fractions in each phase as done in [22]. This will allow us to simulation multi-component solids or liquids (i.e., particles and droplets) with fluid-phase chemical reactions. Due to the differences in temperature and species concentrations, we expect that the reaction rates in the bulk-fluid and added-mass phases can be very different. This difference may be important for the prediction of undesirable by-products such as soot in spray combustion. Furthermore, while we have used the model for high-speed gas flows in this work, the added-mass contribution to heat and mass transfer in liquid–liquid and bubbly flows is also expected to be significant due to the large Schmidt number in liquids. Finally, for low-speed flows the diffusive transport of chemical species and thermal energy by pseudoturbulence in the fluid phase can be significant, and should be included in the two-fluid model [32]. Since the 11-equation model already includes the PTKE, adding pseudo-turbulent diffusion and viscosity should be relatively simple.

CRediT authorship contribution statement

Victor Boniou: Writing – review & editing, Writing – original draft, Software. **Rodney O. Fox:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Jacob W. Posey:** Writing – review & editing, Writing – original draft, Conceptualization. **Ryan W. Houim:** Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The multiphase-modeling approach [21] developed further in this work was initiated by ROF while visiting Professor Marin's group at Ghent University as the International Francqui Chair 2016–2017.

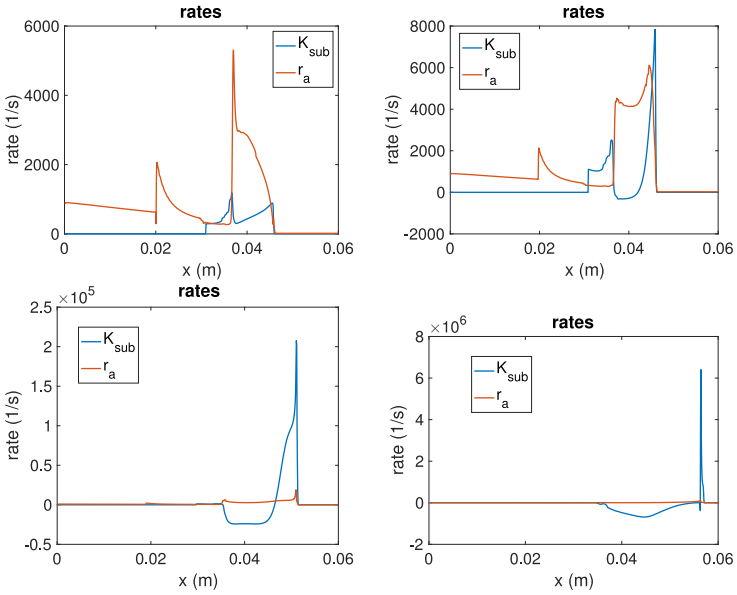


Fig. 9. Rates for sublimation and added-mass exchange with $C_a = 0.001$ at $t = 0.1$ ms. Top left: $K_0 = 1$. Top right: $K_0 = 10$. Bottom left: $K_0 = 100$. Bottom right: $K_0 = 1000$. Negative K_{sub} corresponds to condensation.

Appendix A. Extension to multiple chemical species with homogeneous reactions

In order to model homogeneous chemical reactions, the conserved variables must include the mass fractions of the chemical species. The species balances for the mass fractions in the three phases are [22]

$$\partial_t(\alpha_p\rho_p Y_{p,j}) + \partial_{\mathbf{x}}\cdot(\alpha_p\rho_p Y_{p,j}\mathbf{u}_p) = \alpha_p\dot{\omega}_{p,j} - S^Y_{p,j}, \tag{A.1}$$

$$\partial_t(\alpha_a\rho_a Y_{a,j}) + \partial_{\mathbf{x}}\cdot(\alpha_a\rho_a Y_{a,j}\mathbf{u}_p) = \alpha_a\dot{\omega}_{a,j} + S^Y_{a,j} + S^Y_{p,j} \tag{A.2}$$

and

$$\partial_t(\alpha^\star_f\rho_f Y_{f,j}) + \partial_{\mathbf{x}}\cdot(\alpha^\star_f\rho_f Y_{f,j}\mathbf{u}_f) = \alpha^\star_f\dot{\omega}_{f,j} - S_{a,j}. \tag{A.3}$$

The mass fractions Y_j of species $j = 1,\dots,N$ sum to unity such that

$$S^M = \sum_{j=1}^N S^Y_{p,j} \quad \text{and} \quad S_a = \sum_{j=1}^N S^Y_{a,j}. \tag{A.4}$$

Likewise, the homogeneous reaction rates conserve the elements in each phase such that

$$\sum_{j=1}^N \dot{\omega}_{p,j} = 0, \quad \sum_{j=1}^N \dot{\omega}_{a,j} = 0 \quad \text{and} \quad \sum_{j=1}^N \dot{\omega}_{f,j} = 0. \tag{A.5}$$

Note that because the temperatures and mass fractions in the two fluid phases can be very different, the reaction rates $\dot{\omega}_{a,j}$ and $\dot{\omega}_{f,j}$ can also be very different. By convention, a chemical species that does not exist in the solid/fluid phase has zero mass fraction in that phase, and $S^Y_{p,j}$ is null for that species. Otherwise, the species-exchange terms $S^Y_{p,j}$ and $S^Y_{a,j}$ have the same form as the mass-exchange terms in Section 3.3 with, for example, ρ_f replaced by $\rho_f Y_{f,j}$. As is usual for high-speed flows [22], the molecular-diffusion fluxes have been neglected in the species balances.[7]

Given the mass fractions, the fluid total energy E_f and the internal energies e_a and e_p are treated as described in [22]. Finally, the fluid pressures p_f and p_a depend on the mass fractions through an equation of state. For example, a key step in the numerical algorithm is to find the fluid densities using (4), which requires as input the density ratio $\kappa = \rho_a/\rho_f$. If the ideal gas equation of state is employed, equal pressures yields

$$\kappa = \frac{T_f M_a}{T_a M_f} \tag{A.6}$$

where the average molecular weights are defined by

$$\frac{1}{M_a} = \sum_{j=1}^N \frac{Y_{a,j}}{M_j} \quad \text{and} \quad \frac{1}{M_f} = \sum_{j=1}^N \frac{Y_{f,j}}{M_j}, \tag{A.7}$$

and M_j is the molecular weight of species j. The temperatures T_a and T_f are found from e_a and E_f, respectively, given thermodynamic data for the enthalpies of formation and specific heats of the chemical species [22].

Appendix B. Numerical treatment of source terms

As is customary for fluid–particle flow codes [22], the source terms are handled using operator splitting. In our simulation code, the source terms for mass transfer involving $\dot{M}$ and added mass are treated separately as follows.

B.1. Mass-transfer source terms

Including only the mass-transfer terms involving $\dot{M}$, the balances can be written as

$$\begin{aligned}
\partial_t(\rho_p\alpha_p) &= -\dot{M}, & \partial_t(\rho_a\alpha_a) &= \dot{M},\\
\partial_t(\rho_p\alpha_p\mathbf{u}_p) &= -\dot{M}, & \partial_t(\rho_a\alpha_a\mathbf{u}_p) &= \dot{M},\\
\partial_t(\rho_p\alpha_p K_p) &= -\dot{M}, & \partial_t(\rho_a\alpha_a K_p) &= \dot{M},\\
\partial_t(\rho_p\alpha_p e_p) &= -\max(\dot{M},0)\,e_p - \min(\dot{M},0)\,e_a, & \partial_t(\rho_a\alpha_a e_a) &= \max(\dot{M},0)\,e_p + \min(\dot{M},0)\,e_a.
\end{aligned} \tag{B.1}$$

Conservation of mass, momentum and energy, respectively, yield the following relations:

$$\begin{aligned}
\partial_t(\rho_p\alpha_p) + \partial_t(\rho_a\alpha_a) = \partial_t(\rho_e\alpha^\star_p) &= 0,\\
\partial_t(\rho_e\alpha^\star_p\mathbf{u}_p) &= 0,\\
\partial_t(\rho_e\alpha^\star_p K_p) &= 0,\\
\partial_t(\rho_p\alpha_p e_p) + \partial_t(\rho_a\alpha_a e_a) &= 0.
\end{aligned} \tag{B.2}$$

[7] For the particle and added-mass phases, species diffusion depending on Θ_p (i.e., due to fluctuating particle motion) should also be included in the species balances. In most cases, this 'kinetic' diffusion will be significantly larger than molecular diffusion in the fluid phase.

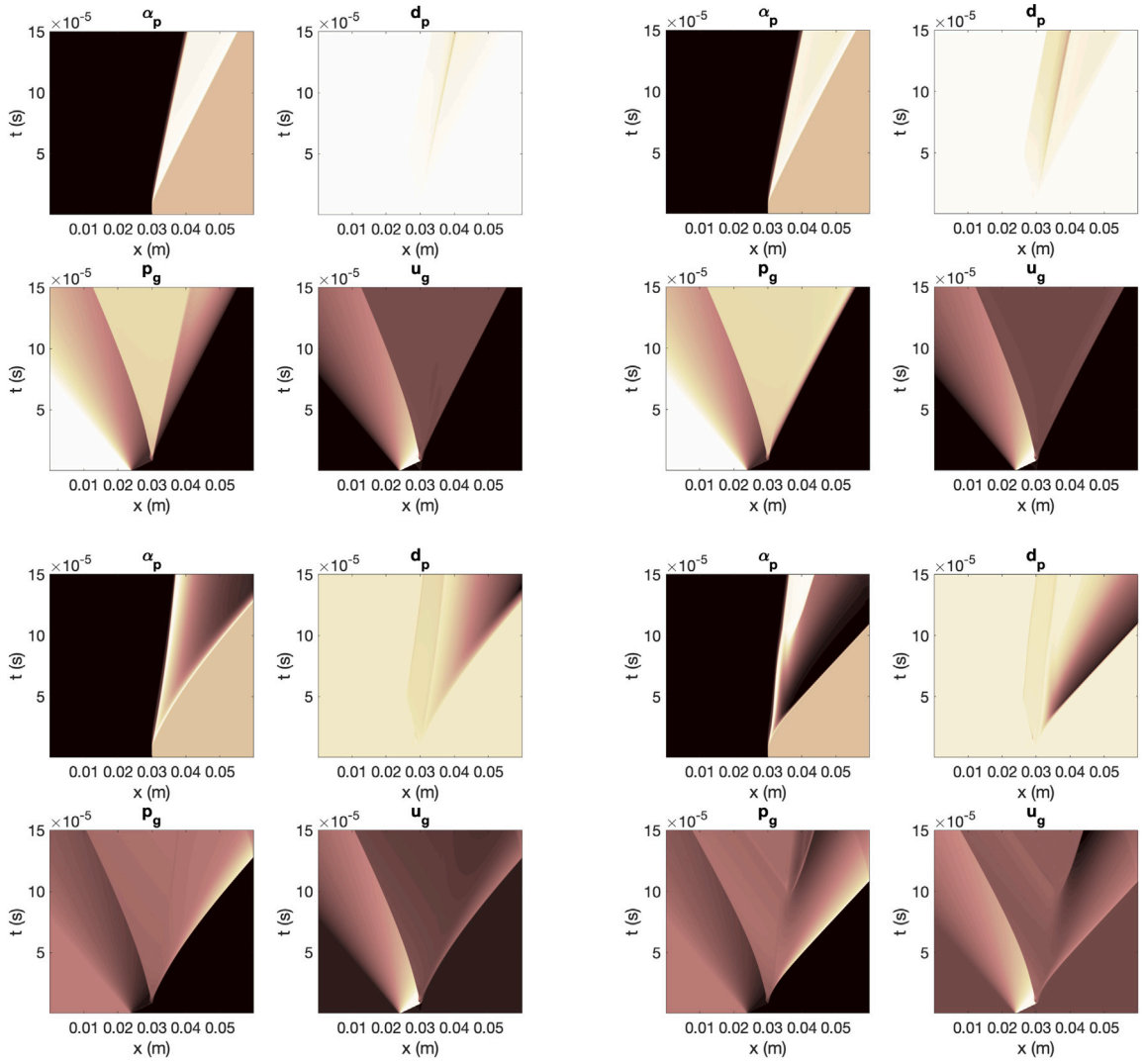


Fig. 10. x–t plots of α_p, d_p, p_g and u_g with increasing sublimation rate. Top left: $K_0 = 1$. Top right: $K_0 = 10$. Bottom left: $K_0 = 100$. Bottom right: $K_0 = 1000$. For large K_0, α_p and d_p drop nearly to zero (black region on right) at the shock location before rising rapidly behind the shock due to condensation (white region). The contours, ranging from low (black) to high (right), are different for each K_0. In order to compare the relative magnitudes for different K_0, the line plots in Figs. 7 and 8 provide the ranges at $t = 0.1$ ms.

These equations can be integrated analytically to give algebraic constraints. For example, the first three imply that $\rho_e \alpha_p^\star$, $\mathbf{u}_p$ and K_p are constant.

In order to ensure that the variables remain physical for an arbitrary time step t, the cases with $\dot{M} < 0$ and $\dot{M} > 0$ are considered separately:

- If $\dot{M} > 0$, then define $C_\xi = K_s$ at the beginning of the time step ($t = 0$). The variables at time $0 < t$ are then

$$\begin{aligned} \rho_p\alpha_p(t) &= \rho_p\alpha_p(0)\, e^{-C_\xi t}, & (\rho_a\alpha_a)(t) &= (\rho_a\alpha_a)(0) + \rho_p\alpha_p(0)[1 - e^{-C_\xi t}], \\ (\rho_p\alpha_p e_p)(t) &= (\rho_p\alpha_p e_p)(0)\, e^{-C_\xi t}, & (\rho_a\alpha_a e_a)(t) &= (\rho_a\alpha_a e_a)(0) + (\rho_p\alpha_p e_p)(0)[1 - e^{-C_\xi t}]. \end{aligned} \tag{B.3}$$

- If $\dot{M} < 0$, then define $C_\xi = -K_s$ at the beginning of the time step ($t = 0$). The variables at time $0 < t$ are then

$$\begin{aligned} (\rho_a\alpha_a)(t) &= (\rho_a\alpha_a)(0)\, e^{-C_\xi t}, & \rho_p\alpha_p(t) &= \rho_p\alpha_p(0) + (\rho_a\alpha_a)(0)[1 - e^{-C_\xi t}], \\ (\rho_a\alpha_a e_a)(t) &= (\rho_a\alpha_a e_a)(0)\, e^{-C_\xi t}, & (\rho_p\alpha_p e_p)(t) &= (\rho_p\alpha_p e_p)(0) + (\rho_a\alpha_a e_a)(0)[1 - e^{-C_\xi t}]. \end{aligned} \tag{B.4}$$

These equations ensure conservation of mass and internal energy for arbitrary t, but will be accurate only for $C_\xi t \ll 1$, i.e., the splitting time step must be suitably small.

B.2. Added-mass source terms

The balance equations with only the added-mass source terms are

$$\begin{aligned} \partial_t(\rho_a\alpha_a) &= S_a, & \partial_t(\rho_f\alpha_f^\star) &= -S_a, \\ \partial_t(\rho_a\alpha_a\mathbf{u}_p) &= \mathbf{S}_\mathbf{u}, & \partial_t(\rho_f\alpha_f^\star\mathbf{u}_f) &= -\mathbf{S}_\mathbf{u}, \\ \partial_t(\rho_a\alpha_a K_p) &= S_K, & \partial_t(\rho_f\alpha_f^\star E_f) &= -S_E, \\ \partial_t(\rho_a\alpha_a e_a) &= S_e, & \partial_t(\rho_f\alpha_f^\star k_f) &= -S_k. \end{aligned} \tag{B.5}$$

From these balances, we observe that conservation of mass, momentum and energy yields, respectively,

$$\begin{aligned} \partial_t(\rho_f\alpha_f^\star) + \partial_t(\rho_a\alpha_a) &= 0, \\ \partial_t(\rho_f\alpha_f^\star\mathbf{u}_f) + \partial_t(\rho_a\alpha_a\mathbf{u}_p) &= 0, \\ \partial_t(\rho_f\alpha_f^\star E_f) + \partial_t(\rho_a\alpha_a K_p) + \partial_t(\rho_a\alpha_a e_a) &= 0; \end{aligned} \tag{B.6}$$

which can be integrated analytically. We refer to the resulting algebraic relations as the conservation constraints. Recall that the PTKE k_f is included in the total energy E_f (and in K_f) and thus does not generate a separate conservation constraint.

For the added-mass source terms, the numerical algorithm is as follows:

1. First compute the change in α_a:

$$\Delta\alpha_a = \alpha_a(t) - \alpha_a(0) = [c_m^\star \alpha_f \alpha_p - \alpha_a(0)][1 - e^{-t/\tau_a}] \tag{B.7}$$

where τ_a is evaluated at $t = 0$. Since α_p is constant, this solution is exact if τ_a is constant.

2. For the particle and added-mass conserved variables, the changes are given by

$$\begin{aligned}
\rho_a\alpha_a(t) &= \rho_a\alpha_a(0) &&+ \begin{cases} \rho_a(0)\Delta\alpha_a & \text{if } \Delta\alpha_a < 0 \\ \rho_f(0)\Delta\alpha_a & \text{if } \Delta\alpha_a > 0, \end{cases} \\
\rho_e\alpha_p^\star \mathbf{u}_p(t) &= \rho_e\alpha_p^\star \mathbf{u}_p(0) &&+ \begin{cases} \rho_a\mathbf{u}_p(0)\Delta\alpha_a & \text{if } \Delta\alpha_a < 0 \\ \rho_f\mathbf{u}_f(0)\Delta\alpha_a & \text{if } \Delta\alpha_a > 0, \end{cases} \\
\rho_e\alpha_p^\star K_p(t) &= \rho_e\alpha_p^\star K_p(0) &&+ \begin{cases} \rho_a K_p(0)\Delta\alpha_a & \text{if } \Delta\alpha_a < 0 \\ \rho_f K_f(0)\Delta\alpha_a & \text{if } \Delta\alpha_a > 0, \end{cases} \\
\rho_a\alpha_a e_a(t) &= \rho_a\alpha_a e_a(0) &&+ \begin{cases} \rho_a e_a(0)\Delta\alpha_a & \text{if } \Delta\alpha_a < 0 \\ \rho_f e_f(0)\Delta\alpha_a & \text{if } \Delta\alpha_a > 0. \end{cases}
\end{aligned} \tag{B.8}$$

The quantities $K_f(0)$ and $e_f(0)$ are found from the relations $K_f = \frac{1}{2}u_f^2 + k_f$ and $e_f = E_f - K_f$.

3. The three conservation constraints provide the changes in the fluid phase:

$$\begin{aligned}
\rho_f\alpha_f^\star(t) &= \rho_f\alpha_f^\star(0) &&- \begin{cases} \rho_a(0)\Delta\alpha_a & \text{if } \Delta\alpha_a < 0 \\ \rho_f(0)\Delta\alpha_a & \text{if } \Delta\alpha_a > 0, \end{cases} \\
\rho_f\alpha_f^\star \mathbf{u}_f(t) &= \rho_f\alpha_f^\star \mathbf{u}_f(0) &&- \begin{cases} \rho_a\mathbf{u}_p(0)\Delta\alpha_a & \text{if } \Delta\alpha_a < 0 \\ \rho_f\mathbf{u}_f(0)\Delta\alpha_a & \text{if } \Delta\alpha_a > 0, \end{cases} \\
\rho_f\alpha_f^\star E_f(t) &= \rho_f\alpha_f^\star E_f(0) &&- \begin{cases} [\rho_a K_p(0) + \rho_a e_a(0)]\Delta\alpha_a & \text{if } \Delta\alpha_a < 0 \\ [\rho_f K_f(0) + \rho_f e_f(0)]\Delta\alpha_a & \text{if } \Delta\alpha_a > 0. \end{cases}
\end{aligned} \tag{B.9}$$

4. The PTKE changes as

$$\rho_f\alpha_f^\star k_f(t) = \rho_f\alpha_f^\star k_f(0) - \begin{cases} \frac{3}{2}\rho_a\Theta_p(0)\Delta\alpha_a & \text{if } \Delta\alpha_a < 0 \\ \rho_f k_f(0)\Delta\alpha_a & \text{if } \Delta\alpha_a > 0 \end{cases}$$

where $\Theta_p(0)$ is found from the relation $3\Theta_p = 2K_p - u_p^2$.

The equations given above ensure conservation of mass, momentum and energy for arbitrary t, but they will be accurate only for constant τ_a. Since the source terms for mass transfer and added mass will balance each other, the largest C_ξ should be used to fix the time step. In the shock-tube problem we implicitly assume that the time step based on the CFL condition is smaller than those from the source terms. However, this should be verified for other applications.

Data availability

No data was used for the research described in the article.

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